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by Hasso-Plattner-Institut

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First Steps Towards Your Own AI Application

Setting up the key tools for local AI development



Design IT.
Create Knowledge.

www.hpi.de

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Attendance list

- Please sign the attendance list and pass it on

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Let's get to know each other

- What is your background?
- What is your motivation coming here today?
- Are you familiar with VSCode, Git, Docker, Python, … ?

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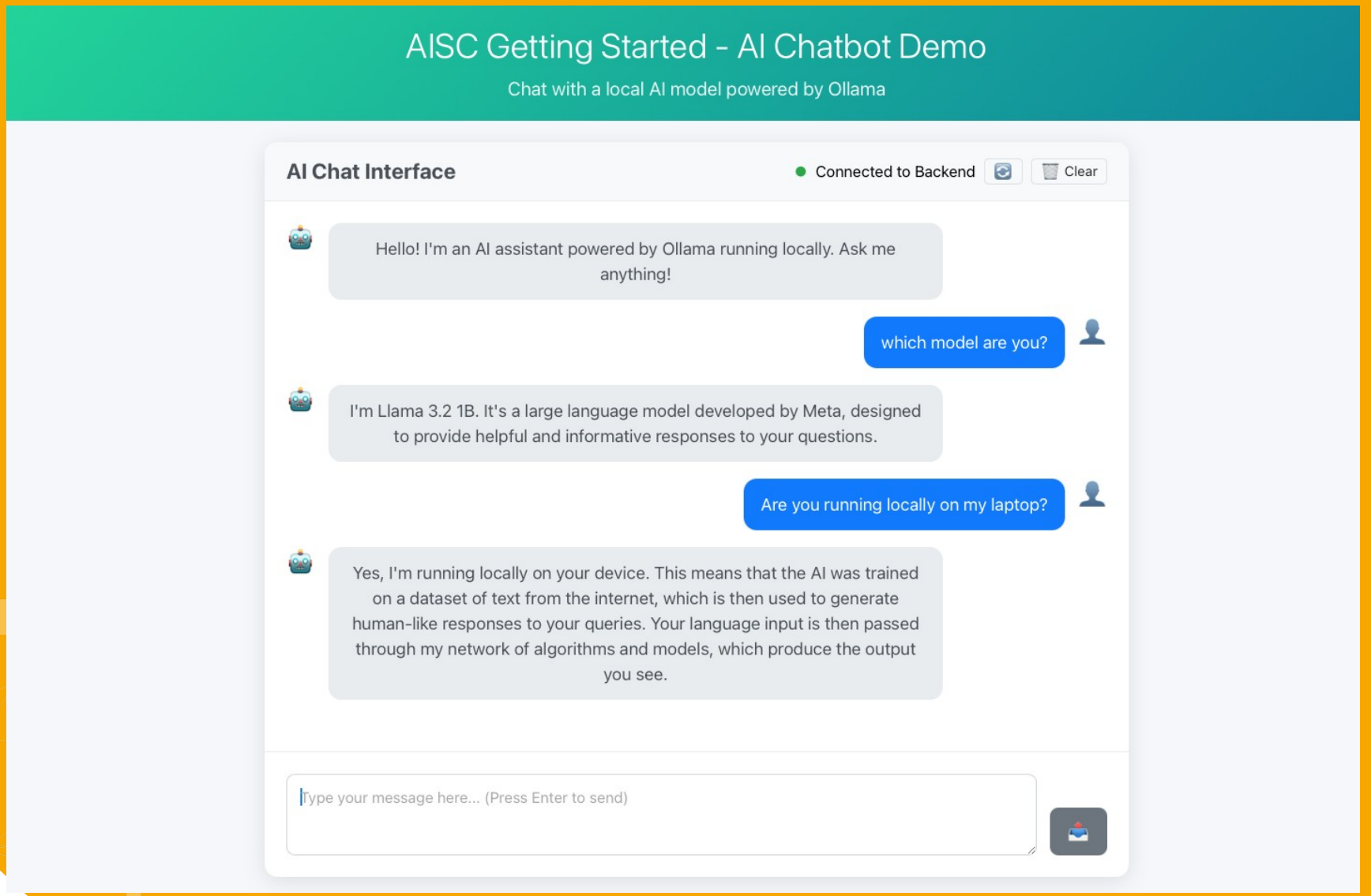
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Goal: Set up your environment

1. VS Code
2. WSL
3. Git
4. Python
5. UV
6. Ollama
7. Docker Desktop
8. Example Application: Local Chatbot

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and Space

Overview of the software



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Visual Studio Code (VS Code)

- Code-Editor:
  - Lightweight
  - Cross-platform
  - Expandable with extensions
- Supports various tools:
  - Python, Jupyter Notebook, Git, Docker, SSH
- Integration of coding assistants (e.g., Claude)



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GitHub & Git

- Platform for version control of code
  - Development, version control, review, and documentation



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Python

- General-purpose programming language with key libraries for AI and data science
  - NumPy: numerical computing
  - PyTorch: neural networks
  - Librosa: audio processing
  - ...
- Extensive learning resources and tutorials available
- Suitable for data processing, model training, ...



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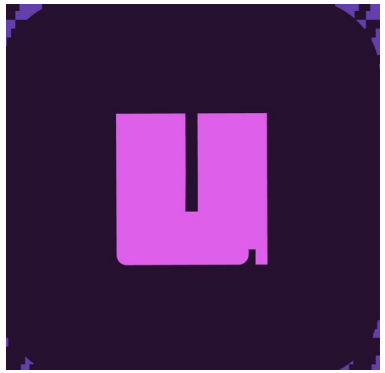
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UV

- Fast Python package and project manager
  - Handles package installation and dependency management
  - Creates reproducible development environments, ensuring consistent package versions

- Ensures code runs reliably across different machines and operating systems
  - Windows 10, 11, XP
  - Linux Mint, Ubuntu, Debian
  - MacOS

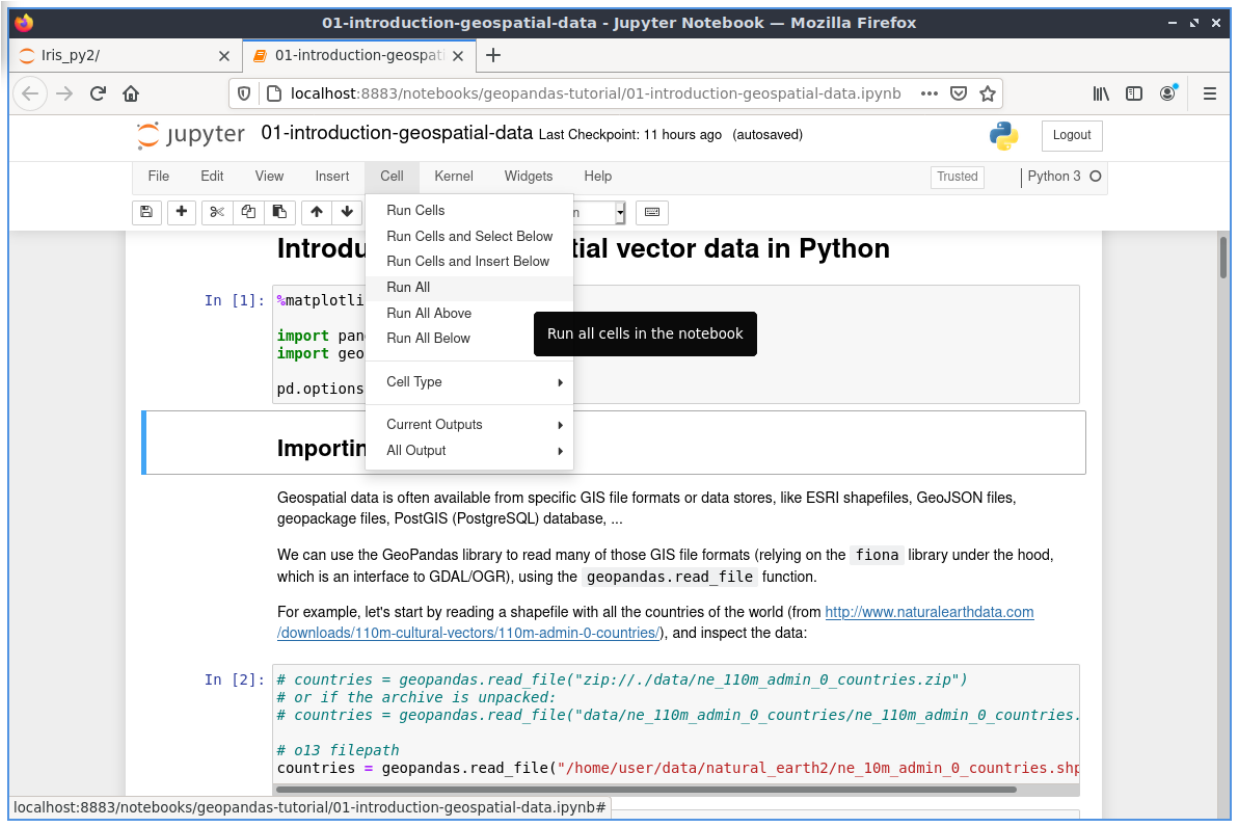


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Jupyter notebooks

- Supports code, narrative text, equations (Markdown), and visualizations
- Ideal for exploratory data analysis, prototyping, and teaching
- Code cells can be executed independently --> Easy debugging

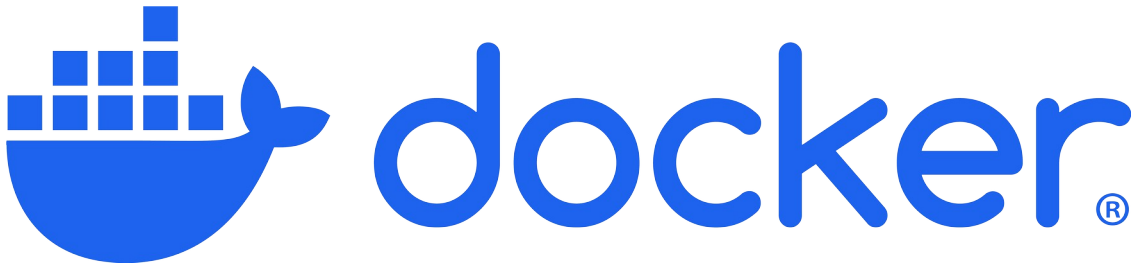


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Docker

- Container platform for packaging reproducible runtime environments
  - Enables applications to run consistently across machines, servers, and high-performance clusters (HPC)
  - Solves "works on my machine" problem
- Ideal for complex software stacks



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Please group together with your Operating System

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Bundesministerium für Forschung, Technologie und Raumfahrt

MacOS?

-> Please sit to the left

Linux?

-> Please sit in the middle

Windows?

-> Please sit to the right

Hands-on

https://github.com/aihpi/workshop-getting-started

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AI for Everybody — Getting Started

This repository hosts the **Getting Started** workshop offered by the AI Service Centre Berlin Brandenburg. It introduces the development tools and workflows — VSCode, Git, WSL, Python, UV, Ollama, and Docker — that participants will use across our workshops.

Workshop materials

All workshop content lives in `03_workshop/`. The workshop's own README is the entry point:

`03_workshop/README.md`

It contains the agenda, the per-tool guides, the Jupyter notebooks for replay at home, and the slides.

Example application

This repository also includes a small chatbot application that demonstrates what you can build once your environment is set up:

- `01_frontend/` — React chat interface
- `02_backend/` — FastAPI server with Ollama integration
- `docker-compose.yml` — runs the full stack locally

The chatbot is the final demo of the workshop (section 8). See `03_workshop/guides/08_chatbot_demo.md` for the walkthrough.

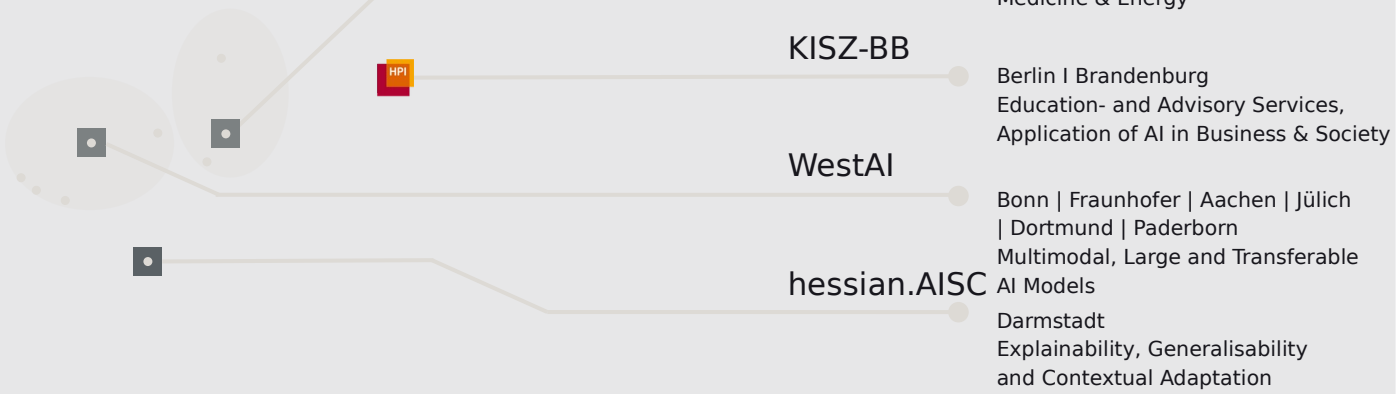
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Overview of the AI Service Center

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AI-Service Centers



KISSKI

Hannover I Göttingen I Kassel
Sensible and Critical Infrastructure
Medicine & Energy

KISZ-BB

Berlin I Brandenburg
Education- and Advisory Services,
Application of AI in Business & Society

WestAI

Bonn | Fraunhofer | Aachen | Jülich
| Dortmund | Paderborn
Multimodal, Large and Transferable
AI Models

hessian.AISC

Darmstadt
Explainability, Generalisability
and Contextual Adaptation

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ADVISORY

AI Advisory

- General Questions regarding AI,
- Computing Infrastructure,
- Model, Architecture, Framework Selection

AI Pilot Projects (3 months)

1. Understanding and Identifying the Use Case
2. Specifying the Use Case
3. Developing a Solution
4. Finalizing and Documenting

Cooperations

- Co-organized Network Meetings or Workshops

Start-up Support

- All of the above
- Support with Networking

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Infrastructure



Training
- 8 x 8 NVIDIA H100 GPU

Inference
- 5 x 8 NVIDIA A30 GPU

ARM Server
- Ampere Altra Max M128-30 CPU
- 2 x NVIDIA L40 GPUs

GPU Server
- AMD Epyc CPU
- 8 x NVIDIA L40S GPU

Edge
- ARMv8 CPU
- NVIDIA Jetson AGX Module

Storage
- 1.5 PB NVRAM

Network
- 400 Gb/s Infiniband
- 200 Gb/s Ethernet

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EDUCATION & TRAINING

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Talks



Guest Talk: Automatische Deepfake-Erkennung mit KI

Di. 17. März, 17:00

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Kostenlos

MOOCs

tele-TASK by Hasso-Plattner-Institut

OPEN HPI



Chatbots leicht gemacht: Große Sprachmodelle einfach nutzen

SELF-PACED SINCE DEC 10, 2025

DATA SCIENCE

RECORD OF ACHIEVEMENT

DE

Work shops

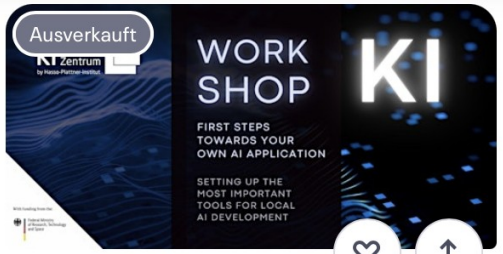


Reinforcement Learning – Part II

Do. 26. Feb., 10:00

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First Steps Towards Your Own AI Application

Donnerstag um 15:00

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Effizientes Abfragen von Informationen – Teil II

Mi. 25. Feb., 13:00

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Reinforcement Learning – Part II

Do. 26. Feb., 10:00

Hasso-Plattner-Institut

Kostenlos



Erste Schritte zu Ihrer eigenen KI-Anwendung

Fr. 13. März, 13:00

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Guest Talk: Automatische Deepfake-Erkennung mit KI

Di. 17. März, 17:00

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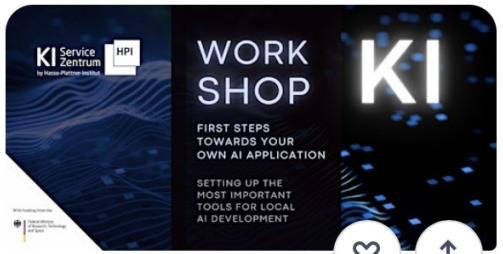


Effizientes Abfragen von Informationen – Teil III

Do. 26. März, 14:00

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First Steps Towards Your Own AI Application

Mo. 13. Apr., 10:00

Hasso-Plattner-Institut

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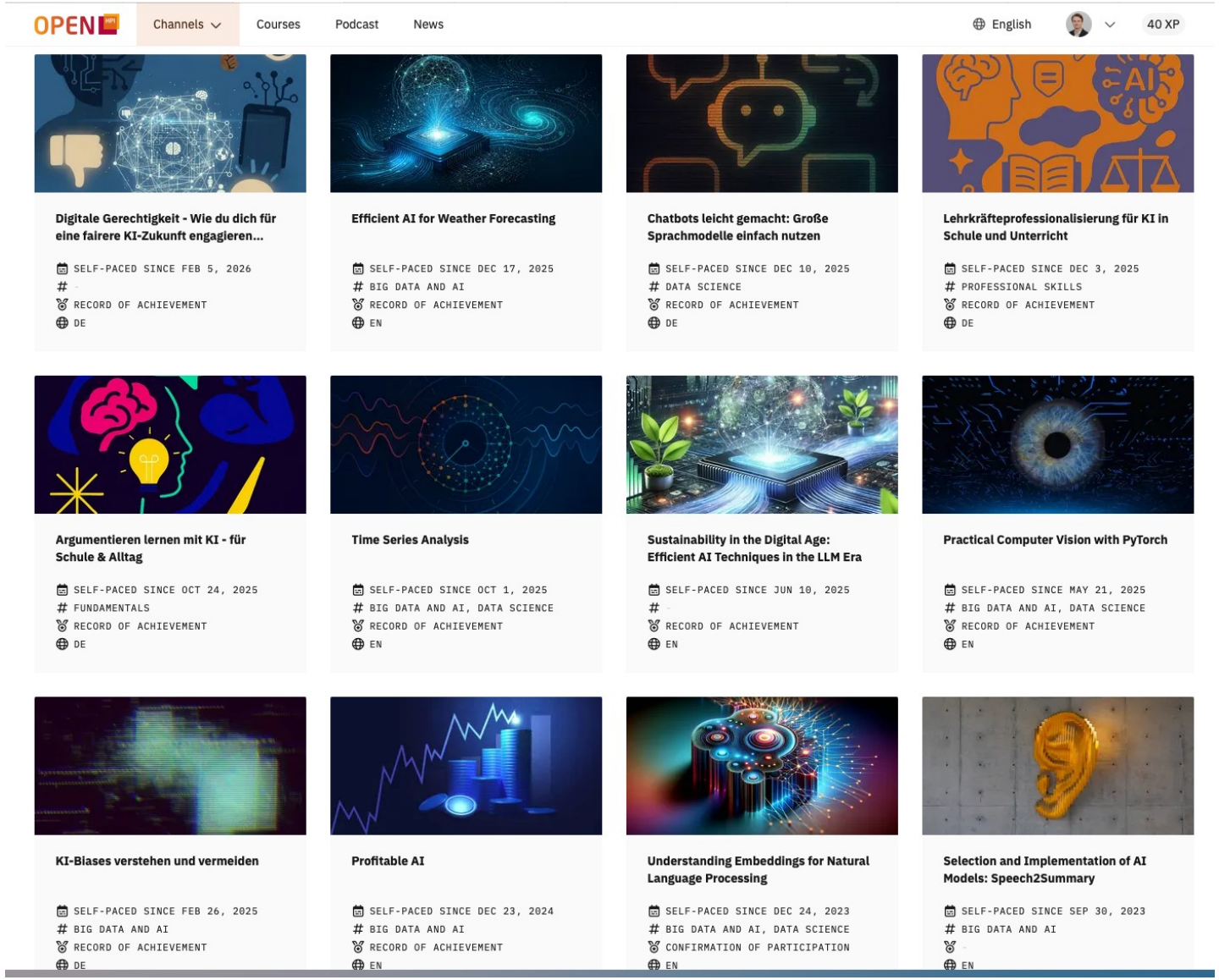


Effizientes Abfragen von Informationen – Teil IV

Di. 28. Apr., 15:00

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News
English
40 XP
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Efficient AI for Weather Forecasting
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BIG DATA AND AI
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EN
Chatbots leicht gemacht: Große Sprachmodelle einfach nutzen
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DATA SCIENCE
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DE
Lehrkräfteprofessionalisierung für KI in Schule und Unterricht
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PROFESSIONAL SKILLS
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Argumentieren lernen mit KI - für Schule & Alltag
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DE
Time Series Analysis
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Sustainability in the Digital Age: Efficient AI Techniques in the LLM Era
SELF-PACED SINCE JUN 10, 2025
RECORD OF ACHIEVEMENT
EN
Practical Computer Vision with PyTorch
SELF-PACED SINCE MAY 21, 2025
BIG DATA AND AI, DATA SCIENCE
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KI-Biases verstehen und vermeiden
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Profitable AI
SELF-PACED SINCE DEC 23, 2024
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Understanding Embeddings for Natural Language Processing
SELF-PACED SINCE DEC 24, 2023
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CONFIRMATION OF PARTICIPATION
EN
Selection and Implementation of AI Models: Speech2Summary
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BIG DATA AND AI
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Your opinion is important!

QR code for the feedback form



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